NUDM Intern Assessment 2026

Property Tax Analytics Dashboard — UPYOG Multi-Tenant Platform

Overview

You are required to build a Property Tax Analytics Dashboard for the UPYOG multi-tenant platform. The platform serves 10 Indian cities, each acting as an independent tenant. A dataset of 1,000 property records is provided as a JSON file. Your task is to load this data in React, display meaningful KPIs, and integrate an AI chat assistant that can answer questions about the data.

Duration	48 Hours
Frontend	React (mandatory)
Backend	Not required — load JSON directly in React
AI Layer	Any free API: Gemini, Claude, or OpenAI
Submission	GitHub repository link sent to HR

The Data File (properties.json)

A file named properties.json is provided alongside this document. It contains exactly 1,000 property records spread across 10 Indian cities. You do not need any database or server — place the file inside your React project and import it directly.

Loading the file in React:

Place properties.json in your src/ folder and import it:

```
import data from './properties.json';
```

Or fetch it from the public/ folder:

```
const res = await fetch('/properties.json'); const data = await res.json();
```

Each record in the file contains the following fields:

Field	Type	Example	Description
property_id	string	UPYOG-DEL-0001	Unique identifier for the property
tenant	string	Delhi	City name — 10 possible values
owner_name	string	Ramesh Kumar	Name of the property owner
property_type	string	Residential	Residential / Commercial / Industrial
ward	string	Ward A	Administrative ward within the city

area_sqft	number	2400	Built-up area in square feet
status	string	Approved	Approved / Rejected / Pending
annual_tax_inr	number	18450.50	Annual property tax in Indian Rupees
collection_inr	number	18450.50	Amount collected (0 if status is not Approved)
registration_date	string	2022-03-15	Date of registration (YYYY-MM-DD)
floor_count	number	2	Number of floors in the building
address	string	42, Sector 12, Delhi	Street address of the property

The 10 cities (tenants) in the dataset are:

Delhi, Mumbai, Pune, Bengaluru, Chennai, Hyderabad, Ahmedabad, Kolkata, Jaipur, Lucknow

Tasks

Task 1 — KPI Dashboard (30 points)

Display the following four KPI cards on the dashboard. All values must update live when the user selects a different city from the tenant filter dropdown.

KPI	How to calculate	Points
Total Properties Registered	Count all records for the selected tenant	7
Total Properties Approved	Count records where status is Approved	7
Total Properties Rejected	Count records where status is Rejected	7
Total Collection (Rs.)	Sum of collection_inr for the selected tenant	9

Tenant filter: A dropdown listing all 10 cities plus an All Cities option. Selecting a city must filter the entire dashboard. This filter is worth 15 additional points.

Task 2 — Comparison Chart (10 points)

Include at least one chart that shows all 10 cities side by side. Recommended libraries: Recharts or Chart.js. Suggested chart types:

- Bar chart showing total collection per city (recommended)

- Pie or donut chart showing share of properties per city

- Grouped bar chart showing Approved vs Rejected vs Pending per city (bonus)

Task 3 — AI Chat Assistant (25 points)

Add a chat box to your dashboard. The user types a question in plain English and receives an answer about the property data. You do not need to train any model — use a free API.

Approach:

1. Compute a summary of the data (totals per city, top collector, etc.).

2. When the user asks a question, send it to the AI API along with the summary as context.

3. Display the AI response in a chat-style UI.

API	Free tier	URL
Google Gemini	Generous free quota (gemini-1.5-flash)	aistudio.google.com
Anthropic Claude	Free trial credits	console.anthropic.com
OpenAI	Free trial credits on new accounts	platform.openai.com

Sample questions the AI should answer correctly:

- Which city has the highest total collection?

- How many properties are rejected in Mumbai?

- What percentage of Delhi properties are approved?

- Which city has the most pending properties?

- Compare total registrations between Pune and Jaipur.

Scoring

Component	Max Points
KPI dashboard (4 cards, correct values)	30
Tenant filter (dropdown updates all KPIs)	15
Comparison chart	10
AI chat assistant	25
Code quality and structure	10
README and setup instructions	10
Bonus — PostgreSQL or ElasticSearch backend	+10
Total	**100 (+10)**

Shortlisting cutoff:

70 and above	Shortlisted for interview
55 to 69	Manual review by hiring team
Below 55	Not shortlisted
No submission or project does not run	Automatically rejected

Submission Instructions

1. Push your completed project to a public GitHub repository.

2. Make sure properties.json is committed and your README.md includes setup steps.

3. Send your GitHub link and Screenshots to [insert HR email] with subject line:

   NUDM Intern Assessment 2025 — [Your Full Name]

4. Deadline: [insert deadline]. Late submissions will not be reviewed.

Important: Store your AI API key in a .env file and add .env to .gitignore. Never commit API keys to GitHub.

Questions? Email [insert contact email]. We will respond within 24 hours.

Good luck. We look forward to seeing what you build.

— NUDM, UPYOG Platform